

AS model refers to the inventory-aware control framework by Marco Avellaneda & Sasha Stoikov (2008), I had zero knowledge about serious market making before this whole paper.

- Description of the market-making strategy

4 strategies, AS_3 is the focus:

1. Baseline: spread calculated by AS model places around mid-quote symmetrically

as for the "frozen inventory" problem from section 2.2. We then set a bid/ask spread given by

formula of the spread from original AS paper:

$$\delta^a + \delta^b = \gamma \sigma^2 (T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right) \quad (3.18)$$

2. AS_1: spread calculated by AS model placed around AS model reservation price symmetrically, T-t replaced by constant tau (due to the nature of ETH perpetual that there is no market close), sigma and k are estimated by EWMA.

formula of the reservation price from original AS paper:

whose solution is $\theta^2 = \frac{1}{2} \sigma^2 \gamma (T - t)$. Thus, for this linear approximation of the order arrival term, we obtain the same indifference price

$$r(s, t) = s - q \gamma \sigma^2 (T - t) \quad (3.17)$$

P.S. A sanity-check simulation code at the beginning is replicating the paper result to ensure this base AS model is implemented correctly in Python (lower variance of PnL and inventory compared to baseline + lower variance of PnL and inventory when risk aversion parameter gamma is increased). and credit to <https://github.com/thibault-charbonnier/market-making-engine> (I translated this C++ to Python)

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=== Monte Carlo Results Summary ===
Total parameter sets tested: 3
Simulations per set: 10000

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	gamma = 0.001		gamma = 0.01		gamma = 0.1	
Metric	AS	Base	AS	Base	AS	Base
Mean PnL	490.2328	490.6082	487.2857	490.5812	454.5683	487.5258
Std PnL	88.6167	92.6694	85.0546	92.3581	76.1329	83.6224
Sharpe Ratio	5.5321	5.2942	5.7291	5.3117	5.9707	5.8301
Mean Final Inv	0.0478	-0.2363	0.1460	-0.4948	0.0946	0.1761
Std Final Inv	18.1240	26.8998	10.3207	26.5706	5.8478	25.6757
Mean Spread	1.3347	1.3347	1.3469	1.3469	1.4706	1.4706

AS - Baseline	gamma = 0.001		gamma = 0.01		gamma = 0.1	
Mean PnL	0.3751	0.3751	-3.2955	-3.2955	-32.0575	-32.0575
Std PnL	-4.0527	-4.0527	-7.3035	-7.3035	-7.4895	-7.4895
Sharpe Ratio	+0.2379	+0.2379	+0.4174	+0.4174	+0.1406	+0.1406
Mean Final Inv	+0.2841	+0.2841	+0.6408	+0.6408	-0.0815	-0.0815
Std Final Inv	-8.7758	-8.7758	-16.2498	-16.2498	-19.8279	-19.8279
Mean Spread	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000

3. AS_2: Based on AS_1, added funding cost skew (current mid quote * -funding rate*a linearly increasing fraction from 1 second / 28800 seconds i.e.8 hours, increment of 1 second) to the reservation price to better reflect the economics of ETH perpetual. Added transaction cost skew (deducted from bid, added on ask).

4. AS_3 (The final model): It is well known that the AS model underperforms in a trending market, and this is visually seems that the asset exhibits some trend. Based on AS_2, I researched on paper “High-frequency market-making with inventory constraints and directional bets by Pietro Fodra, Mauricio Labadie“, added ABM drift rate skew (estimated using OLS regression on order flow imbalance) to better capture the trend in price time series, this is the only directional bet (alpha component) in this whole work.

• If $S(t)$ is an arithmetic Brownian motion with drift, i.e.

$$dS(t) = bdt + \sigma dW(t)$$

then

$$\mathbb{E}_{t,s}[S(T)] = s + b(T - t),$$

which implies that

$$\theta_1 = s + b(T - t), \quad \theta_2 = -\eta - \frac{1}{2}\gamma\sigma^2(T - t).$$

In consequence, the optimal controls are

$$\begin{aligned} \delta_s^\pm &= \frac{1}{\gamma} \log\left(1 + \frac{\gamma}{k}\right) + \eta + \frac{1}{2}\gamma\sigma^2(T - t) \pm \left(b(T - t) - q[2\eta + \gamma\sigma^2(T - t)]\right), \\ \psi_s &= \frac{2}{\gamma} \log\left(1 + \frac{\gamma}{k}\right) + 2\eta + \gamma\sigma^2(T - t), \\ r_s &= s + b(T - t) - q(2\eta + \gamma\sigma^2(T - t)). \end{aligned}$$

In particular, if $b = \eta = 0$ we recover the results of Avellaneda and Stoikov [1].

- Backtesting simulator, including:
 - Order fills

Event driven, queue / execution model that simulates the dynamics in the LOB and realized transaction. Order placing is defaulted to 1 unit every time with max inventory limit. Order cancellation uses a “keep and adjust” logic, it only cancels and replaces when the desired bid / ask next time stamp moves outside a tick-based tolerance. I ensured there is no data look-ahead by printing the timestamp of data used. See below for example, LOB should have the latest timestamp.

```
[ETH step 1,500,000]
LOB ts used      : 2026-03-20T01:09:49.543984206
funding ts used  : 2026-03-20T01:09:48.000000000 (f=+1.977e-05, frac since 8h snapshot=0.1455)
trade ts used    : 2026-03-20T01:09:31.453621606
EWMA sigma      : 0.0766 price/sqrt(s) (87,082 buckets closed)
EWMA k          : 13.3298 (30,278 trades seen)
OFI drift mu     : +1.754546e-02 USD/s (beta=+2.065567e-03, 149 calibs, 87,082 buckets closed, skew@horizon=+0.0877 USD)
mid=2151.55
[baseline ] r=2151.55 bid=2151.4 ask=2151.7 | q=+31.349 ETH PnL=-406.61 (real=-395.68 fund=+6.3323 fees=255.19) fills=2539 placed=66844
  live bid: px=2151.4 rem=1.000 queue_ahead=2.324
  live ask: px=2151.7 rem=1.000 queue_ahead=2.323
[AS_1 ] r=2151.56 bid=2151.4 ask=2151.7 | q=-0.873 ETH PnL=-763.98 (real=-501.66 fund=-1.0296 fees=260.81) fills=2556 placed=74671
  live bid: px=2151.4 rem=1.000 queue_ahead=2.324
  live ask: px=2151.7 rem=1.000 queue_ahead=2.323
[AS_2 ] r=2151.55 bid=2151.2 ask=2151.9 | q=-0.888 ETH PnL=-618.42 (real=-517.70 fund=+2.2112 fees=100.07) fills=880 placed=72399
  live bid: px=2151.2 rem=1.000 queue_ahead=3.601
  live ask: px=2151.9 rem=1.000 queue_ahead=2.323
[AS_3 ] r=2151.63 bid=2151.3 ask=2152.0 | q=-0.169 ETH PnL=-841.76 (real=-724.81 fund=+0.8968 fees=116.70) fills=1005 placed=126491
  live bid: px=2151.2 rem=1.000 queue_ahead=5.324
  live ask: px=2152.0 rem=1.000 queue_ahead=4.113
```

Variant comparison (step 1.500.000)

- Inventory tracking

Analytics in Python notebook: Inventory mean (absolute value because we can go short), standard deviation. Sampled every 10 events in LOB since backtest started. No final inventory because it becomes meaningless without market close (no multiple paths to simulate either)

- PnL calculation

Example in Python notebook:

TOTAL PnL

formula : PnL = equity - initial_cash ; equity = cash + q*mid +funding_accrued

period : cumulative since start of backtest

value : -1,463.94 USD (-0.1464% of initial cash)

DAILY PnL (intraday, current day)

formula : $\text{daily_PnL} = \text{equity}(\text{now}) - \text{equity}(\text{day start}) = d(\text{realized}) + d(\text{unrealized}) + d(\text{funding}) - d(\text{fees})$
 period : current UTC calendar day so far
 value : -264.34 USD (day-start equity 998,800.41)
 breakdown: realized_trading=-171.03 unrealized=-4.39 funding=-1.3237 fees=-87.59
 closed days (end-start): 2026-03-19:-706 2026-03-20:-494

- Performance metrics for the evaluation period
 Total PnL, Sharpe (Risk Adjusted return) and Maximum Drawdown, PnL standard deviation per minute
 For more analytics, please look at the python notebook.
- Brief conclusions and analysis - see below

```
EDGE + ADVERSE-SELECTION BY SIDE (running)
formula : edge = half-spread captured (bid: mid-price, ask: price-mid) ; adverse@H = sgn*(mid(t+H)-mid_fill) ;
net@H = edge + adverse (realised markout PnL per fill, bp of fill-time mid)
reading : edge>0 (we quote around mid) ; adverse<0 = toxic flow ; net>0 = fill profitable by H
bid : edge=+0.789bp (+0.1694 USD/ETH, n=1051) t+1s[adv=-2.686bp/-0.5769USD net=-1.897bp/-0.4075USD] t+3s[adv=-2.718bp/-0.5842USD net=-1.929bp/-0.4148USD] t+5s[adv=-2.621bp/-0.5635USD net=-1.832bp/-0.3941USD]
ask : edge=+0.844bp (+0.1815 USD/ETH, n=1074) t+1s[adv=-2.991bp/-0.6423USD net=-2.147bp/-0.4608USD] t+3s[adv=-3.243bp/-0.6961USD net=-2.399bp/-0.5146USD] t+5s[adv=-3.111bp/-0.6682USD net=-2.267bp/-0.4867USD]
combined: edge=+0.817bp (+0.1755 USD/ETH, n=2125) t+1s[adv=-2.840bp/-0.6100USD net=-2.023bp/-0.4345USD] t+3s[adv=-2.983bp/-0.6407USD net=-2.166bp/-0.4653USD] t+5s[adv=-2.869bp/-0.6164USD net=-2.052bp/-0.4409USD]
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To be frank, it is very difficult to make money no matter what parameters I try. On average my orders are exploited by more informed traders (adverse selection), it's just like a curse that every time my bid gets filled, the mid just falls quickly and every time my ask gets filled, the mid just goes up quickly. My thought process is, essentially my bid and ask are wrong, so why? Maybe the drift rate is too noisy, maybe the parameter k in the AS model is too noisy, maybe the filling model is subtly wrong (engineering side), maybe my modeling is not the right framing of this problem.

Variant	PnL(USD)	Shrp/min	MaxDD	mean q	Fills	Vol(ETH)	transaction cost	Funding
[baseline] Baseline (symmetric mid, AS sprea	-3,648.07	-0.0112	6,138.01	41.5898	6103	2618.533	562.56	+15.0536
[AS_1] AS_1 (funding_skew=False, tc_skew=Fal	-1,628.91	-0.0898	1,661.79	2.4841	6025	2621.437	563.11	-2.8289
[AS_2] AS_2 (funding skew ON, tc skew ON)	-930.60	-0.0344	1,215.71	3.4765	3199	1475.618	316.94	+10.7960
[AS_3] AS_3 (funding skew ON, tc skew ON, dr	-695.71	-0.0266	958.70	3.4377	2266	1166.081	250.17	+4.8931

The only thing I know is, with the inventory aware skew, the bot won't go too crazy to build large position, and the more skew term we have, the more far away our bid / ask are from mid quote and the less orders filled, and hence less bleeding to the PnL under the adverse selection situation, hence the better Sharpe compared to AS_1 and AS2, although it's still a noisy estimation given the amount of data.

